



Conducted By Kalyan Citizens' Education Society

B. K. Birla College of Arts, Science & Commerce, Kalyan
(Empowered Autonomous Status)

Affiliated to University of Mumbai

ISO 9001: 2015 Certified | Reaccredited by NAAC (4th Cycle) with 'A++' Grade (CGPA – 3.51)

B. K. Birla College Campus Road, Kalyan (West) - 421301

CERTIFICATE

This is to certify that **Mr./Ms.** _____

Roll Number/Seat Number _____ a student of

Programme _____ has successfully completed

the practical/project of the **Course** _____

for **Semester** _____ under the valuable guidance of

_____ during the **A.Y.** _____.

Date:

Place:

Internal Examiner

External Examiner

Seal

INDEX

Programme: _____ **Semester:** _____

Course: _____

Sr. No.	Prac. No.	Particulars	Date	Teacher Signature
1.	1.			
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3.	3.			
4.	4.			
5.	5.			
6.	6.			
7.	7.			
8.	8.			

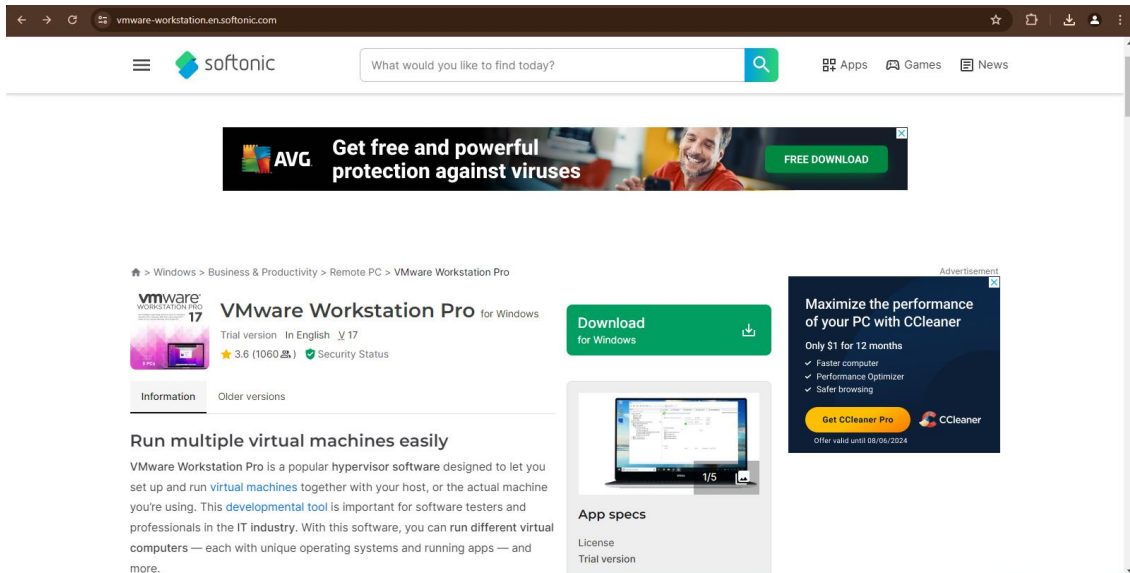
Practical 1

Installing Ubuntu on VMWare Workstation

Theory: To install Ubuntu on VMware Workstation, start by downloading the Ubuntu ISO file from the official Ubuntu website. Open VMware Workstation and select "Create a New Virtual Machine." Choose "Typical" configuration and click "Next." Select "Installer disc image file (iso)" and browse to the downloaded Ubuntu ISO. Continue through the wizard, setting the operating system as Linux and version as Ubuntu. Specify the virtual machine name and location, then allocate disk space (recommended at least 20GB). Follow the prompts to customize hardware settings if needed, then finish the setup and start the virtual machine. The Ubuntu installer will boot, guiding you through the standard installation process to complete the setup.

Step 1: Click on the link provided below. Now click on “Download for windows” and then click on

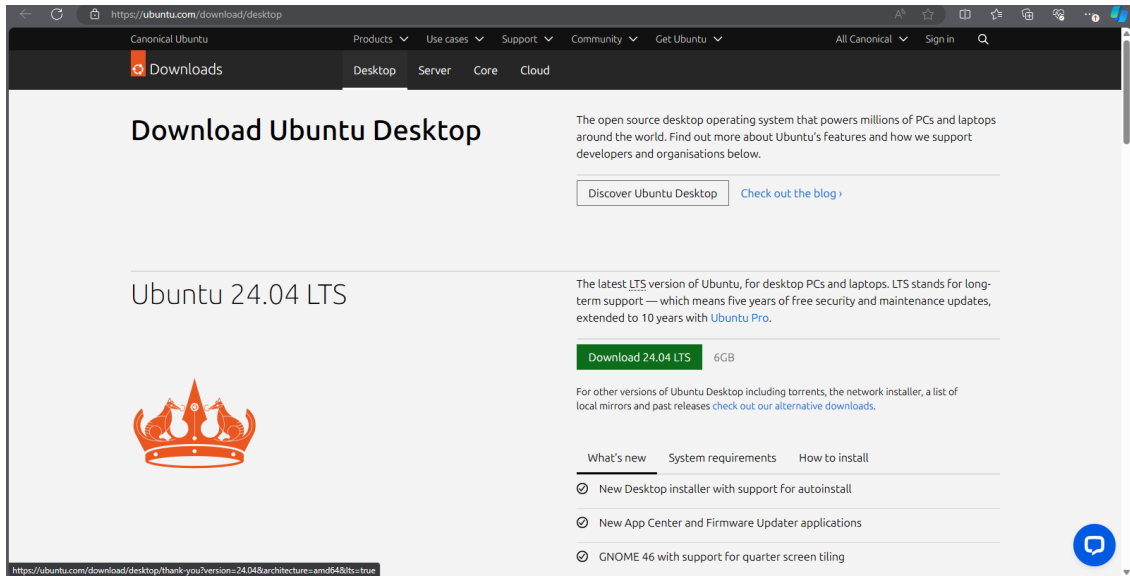
<https://vmware-workstation.en.softonic.com/>



Step 2: Open the (.exe) file and install it.

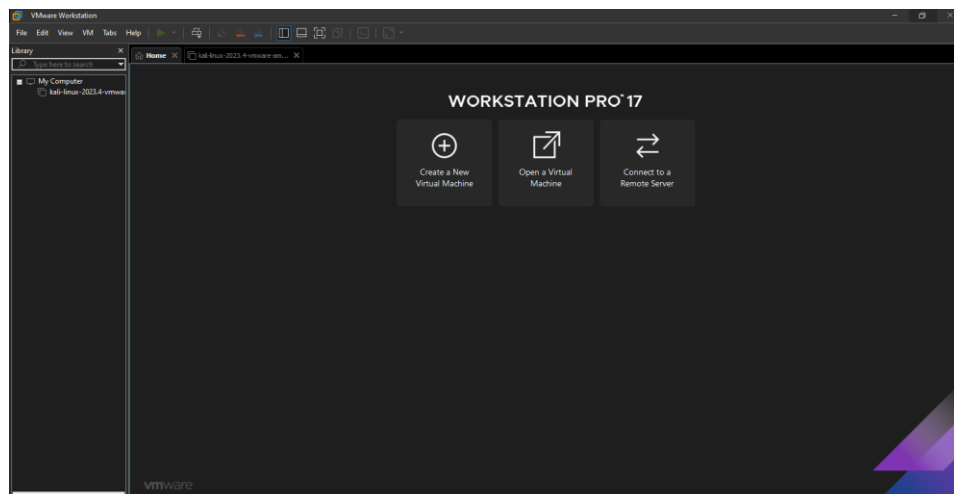
Step 3: Open the link below and download the Ubuntu ISO file.

<https://ubuntu.com/download/desktop>

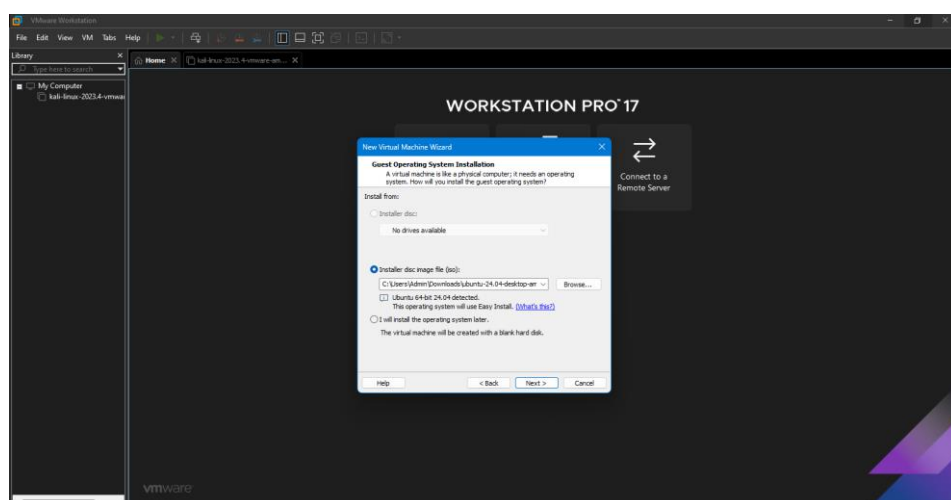


Installing Ubuntu in VMWare

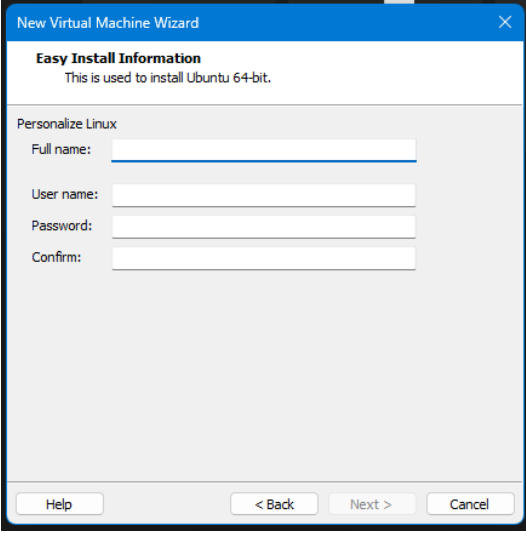
Step 5: Open the installed VMware application.



Step 6: Click on “Create new virtual machine”. Browse the ISO file that has been downloaded and select it.



Step 7: Enter all the details >> click on next. Personalization of Virtual Machine.



New Virtual Machine Wizard

Easy Install Information
This is used to install Ubuntu 64-bit.

Personalize Linux

Full name:

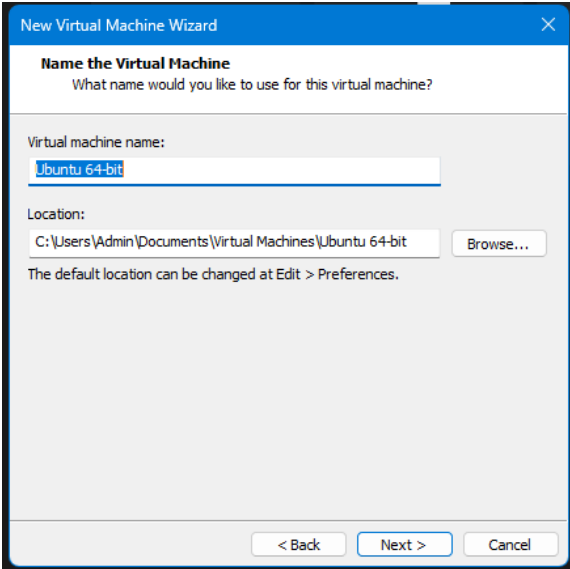
User name:

Password:

Confirm:

Help < Back Next > Cancel

Step 8: Enter Virtual machine name >> click on next.



New Virtual Machine Wizard

Name the Virtual Machine
What name would you like to use for this virtual machine?

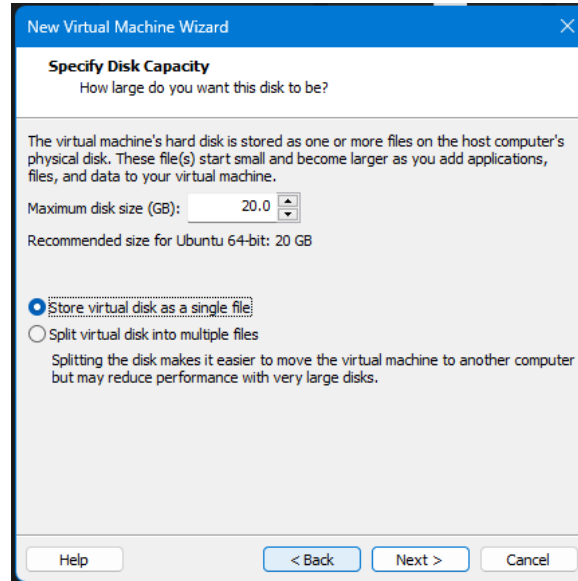
Virtual machine name:

Location:
 Browse...

The default location can be changed at Edit > Preferences.

< Back Next > Cancel

Step 9: Specify disk capacity >> click on next.



The screenshot shows the 'Specify Disk Capacity' step of the 'New Virtual Machine Wizard'. The window title is 'New Virtual Machine Wizard'. The subtitle is 'Specify Disk Capacity' with the question 'How large do you want this disk to be?'. A text box explains: 'The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.' Below this, 'Maximum disk size (GB):' is set to '20.0' with a spin button. It also states 'Recommended size for Ubuntu 64-bit: 20 GB'. There are two radio buttons: 'Store virtual disk as a single file' (selected) and 'Split virtual disk into multiple files'. A note below the second option says: 'Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.' At the bottom are buttons for 'Help', '< Back', 'Next >', and 'Cancel'.

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB): 20.0

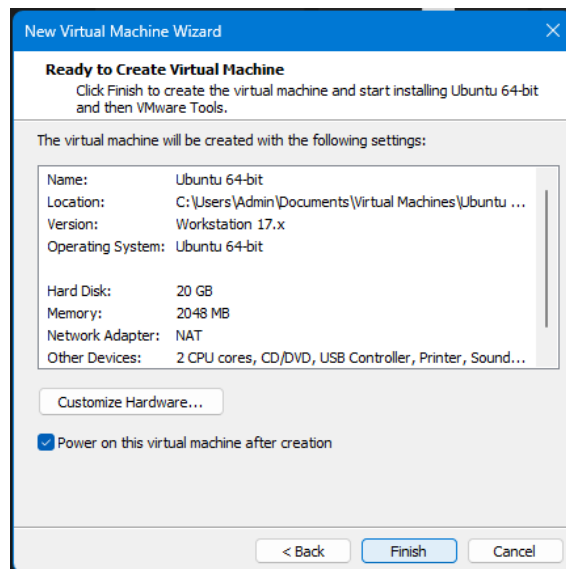
Recommended size for Ubuntu 64-bit: 20 GB

☒ Store virtual disk as a single file
☐ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help < Back Next > Cancel

Step 10: If needed customize the hardware and click on finish.



The screenshot shows the 'Ready to Create Virtual Machine' step of the 'New Virtual Machine Wizard'. The window title is 'New Virtual Machine Wizard'. The subtitle is 'Ready to Create Virtual Machine' with the instruction 'Click Finish to create the virtual machine and start installing Ubuntu 64-bit and then VMware Tools.' A text box lists the settings: 'The virtual machine will be created with the following settings:'. The settings are: Name: Ubuntu 64-bit, Location: C:\Users\Admin\Documents\Virtual Machines\Ubuntu ..., Version: Workstation 17.x, Operating System: Ubuntu 64-bit, Hard Disk: 20 GB, Memory: 2048 MB, Network Adapter: NAT, and Other Devices: 2 CPU cores, CD/DVD, USB Controller, Printer, Sound... Below the list is a 'Customize Hardware...' button. At the bottom, there is a checkbox 'Power on this virtual machine after creation' which is checked. At the very bottom are buttons for '< Back', 'Finish', and 'Cancel'.

Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing Ubuntu 64-bit and then VMware Tools.

The virtual machine will be created with the following settings:

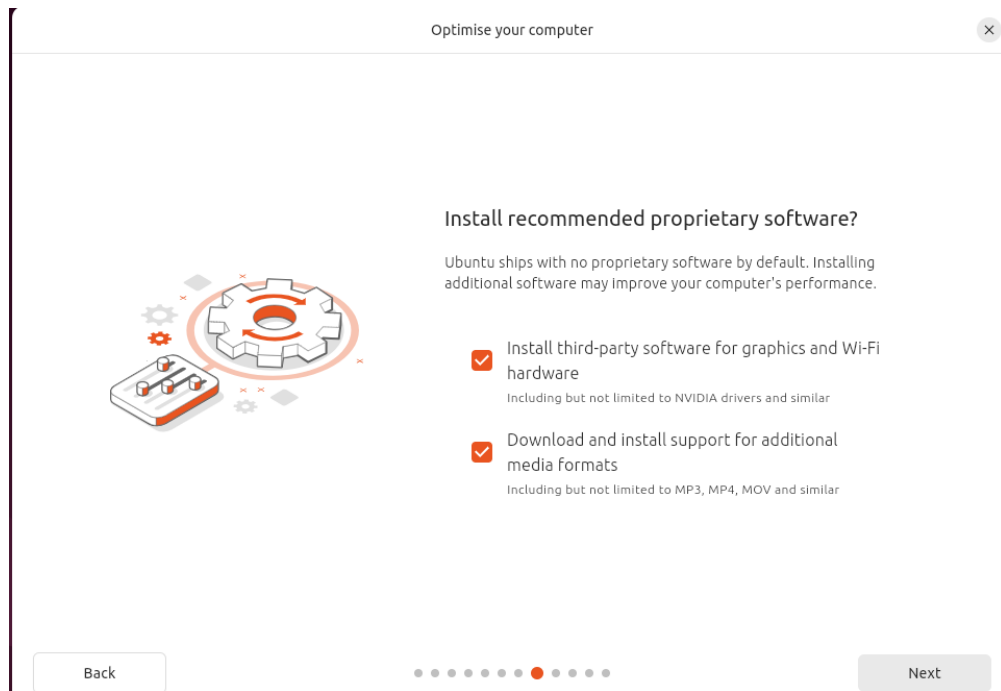
Name:	Ubuntu 64-bit
Location:	C:\Users\Admin\Documents\Virtual Machines\Ubuntu ...
Version:	Workstation 17.x
Operating System:	Ubuntu 64-bit
Hard Disk:	20 GB
Memory:	2048 MB
Network Adapter:	NAT
Other Devices:	2 CPU cores, CD/DVD, USB Controller, Printer, Sound...

Customize Hardware...

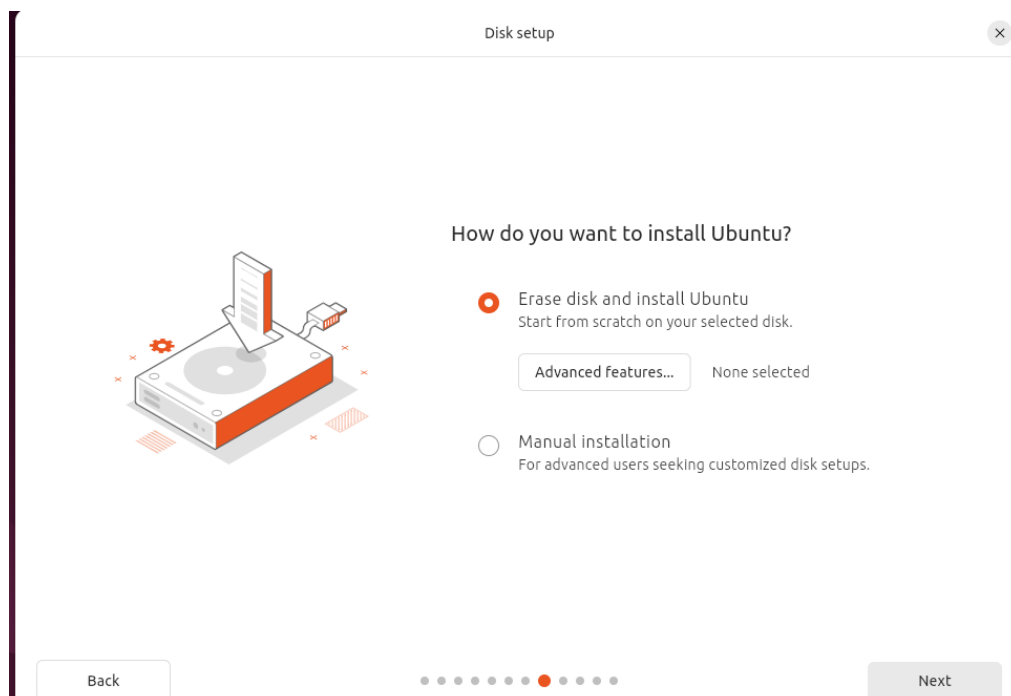
☒ Power on this virtual machine after creation

< Back Finish Cancel

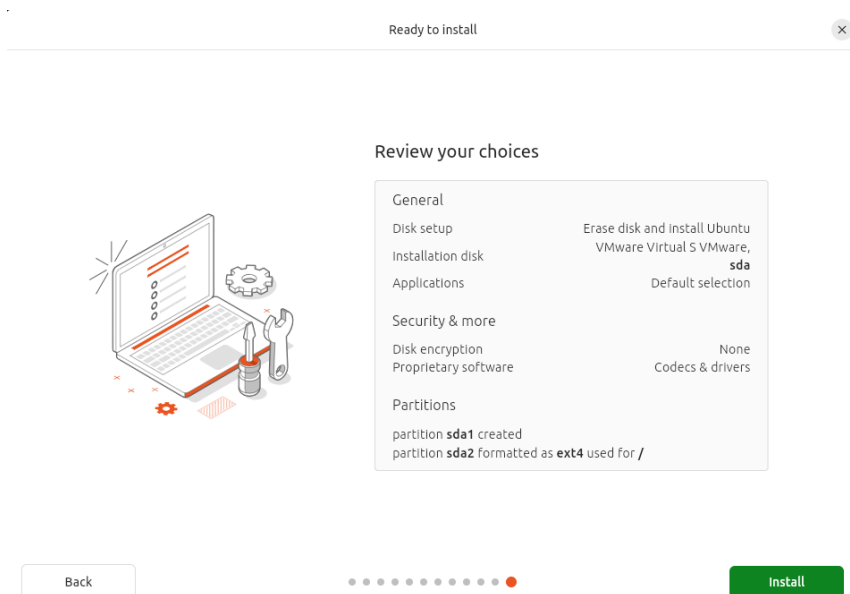
Step 11: Ubuntu has been successfully installed in VMWare. Now open VMware and click on ubuntu-64 which appears on the left panel. (If not directly opened). Select the language and click on next. Select the Normal or Minimal installation based on your requirements.



Step 12: Select Installation type >> Erase disk and install Ubuntu (This will not delete your windows data) >> Click on next..., next....



Step 13: Now click on install.



Practical 2

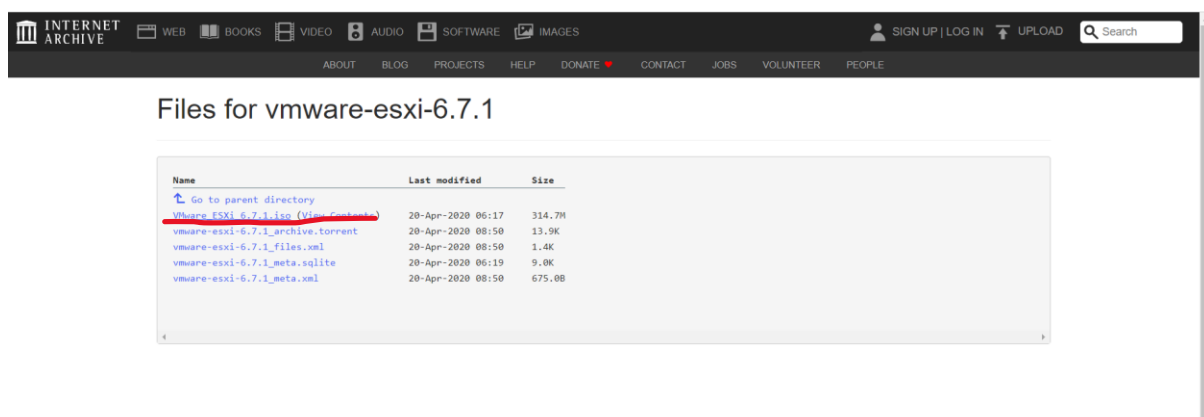
Installing VMware ESXi

Theory: - Installing VMware ESXi involves several steps to set up this bare-metal hypervisor. First, download the ESXi ISO image from the VMware website and create a bootable USB drive or burn the ISO to a CD/DVD. Boot the server from this media, and the ESXi installer will start. Follow the prompts to select the target disk for the ESXi installation, which should be a dedicated drive for the hypervisor. Accept the license agreement, choose the keyboard layout, and set a root password. Once the installation is complete, remove the installation media and reboot the server. After reboot, configure the network settings via the Direct Console User Interface (DCUI) to ensure the server has network connectivity. Finally, access the ESXi management interface (vSphere Client or Web Client) from another machine to manage the ESXi host and begin deploying virtual machines. This process sets up a powerful, virtualized environment for running multiple virtual machines on a single physical server.

Step 1: Prepare Installation Media

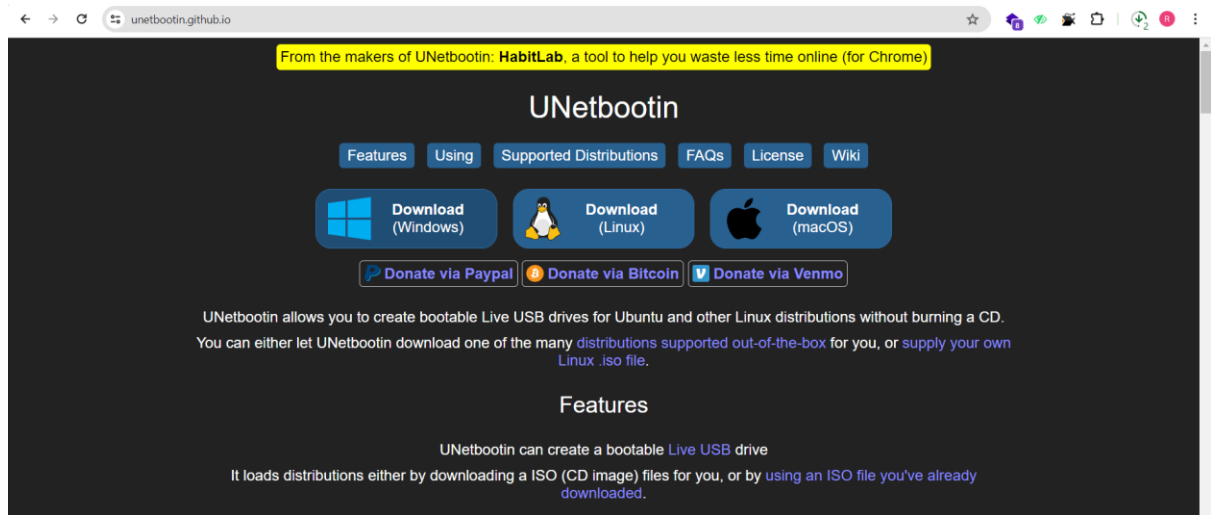
Download the VMware ESXi ISO file

1. Download the latest version of the ESXi ISO file through https://archive.org/download/vmware-esxi-6.7.1/VMware_ESXi_6.7.1.iso.



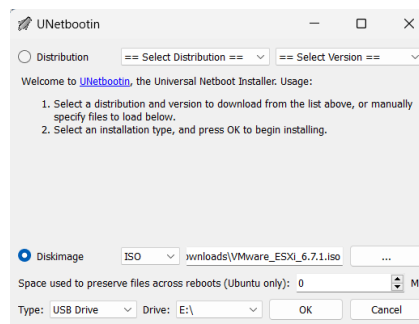
Create a Bootable USB Drive with the ESXi Installer

1. Insert a USB drive into your computer (minimum 8GB recommended).
2. Download and install a tool to create bootable USB drives, such as Rufus(<https://rufus.ie/>) or [UNetbootin](<https://unetbootin.github.io/>).

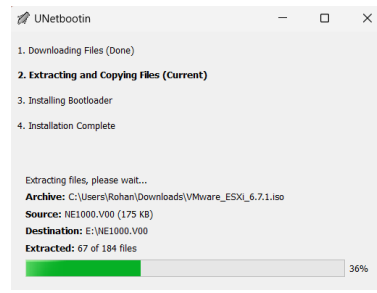


3. Open the tool and select the USB drive you inserted.

4. Choose the downloaded ESXi ISO file as the source.



5. Start the process to create the bootable USB drive.



Step 2: Install ESXi

Boot the Server from the USB Drive

1. Insert the bootable USB drive into the server where you want to install ESXi.
2. Power on or restart the server.
3. Enter the BIOS/UEFI settings (commonly by pressing keys like F2, F10, F12, or Del during boot).
4. Change the boot order to prioritize booting from the USB drive.
5. Save changes and exit the BIOS/UEFI settings. The server should now boot from the USB drive.

Follow the On-Screen Prompts to Complete the Installation

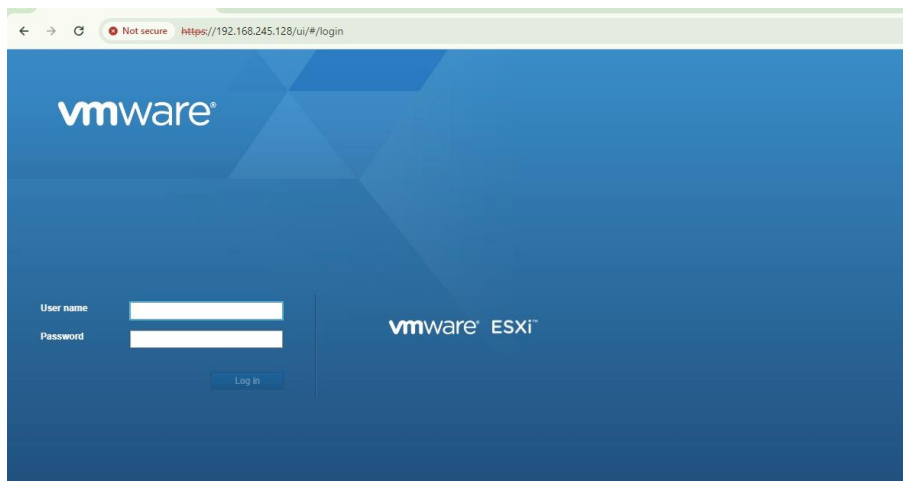
1. When the ESXi installer starts, you'll see a loading screen followed by the Welcome screen.
2. Press **Enter** to continue.
3. Read and accept the End User License Agreement (EULA) by pressing **F11**.
4. Select the disk where you want to install ESXi, and press **Enter**.
5. Confirm the disk selection and press **Enter** to continue.
6. Choose the keyboard layout and press **Enter**.
7. Set a root password for the ESXi host and press **Enter**.
8. Press **F11** to start the installation.

Configure Basic Settings

1. Once the installation is complete, remove the USB drive.
2. Reboot the server.
3. After the server reboots, the Direct Console User Interface (DCUI) will be displayed.
4. Press **F2** to customize the system.
5. Enter the root password you set during installation.
6. Configure the network settings by selecting **Configure Management Network**.
 - Set the IP address, subnet mask, and default gateway.
 - Configure DNS servers if necessary.
7. Adjust the time zone settings under **Configure Management Network > Network Options**.
8. Press **Esc** to exit the configuration menu and apply the changes.

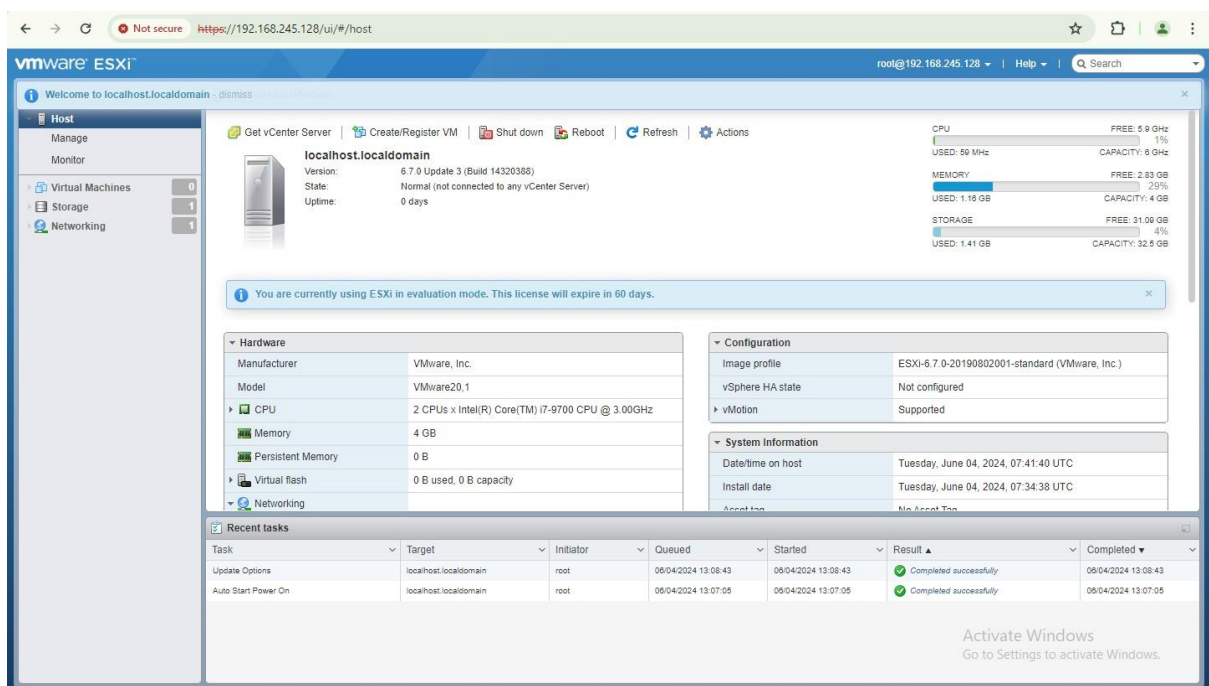
Verification and Access

1. Once the basic configuration is done, you can manage the ESXi server using the VMware Host Client.
2. Open a web browser and navigate to “https://[Your-ESXi-Host-IP-Address]” (https://192.168.245.128).



3. Log in with the root account and password you configured.

You have now successfully installed and configured VMware ESXi on your server.



Practical 3

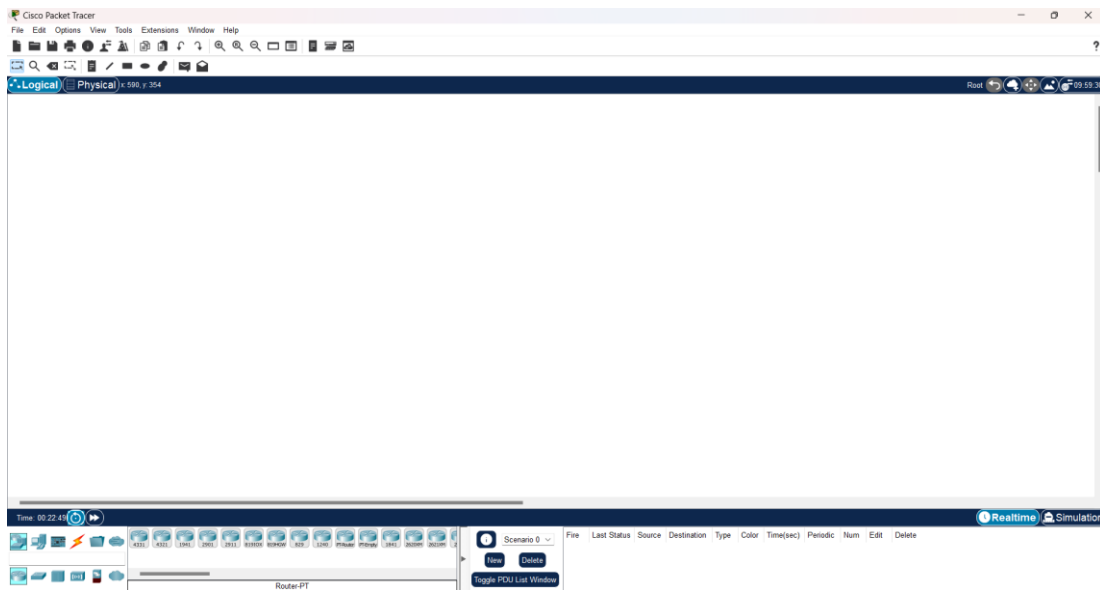
Simulate VLAN Configuration in Packet Tracer

Aim: To understand and configure VLANs for network segmentation and to observe communication between devices within and across VLANs.

Steps to Simulate VLAN Configuration:

1. Open Cisco Packet Tracer:

Launch the Cisco Packet Tracer application.



2. Add Network Devices:

Add a Switch:

From the *Network Devices* section, drag and drop a Switch (e.g., 2960 model) onto the workspace.

Add PCs:

From the *End Devices* section, drag and drop 4 PCs onto the workspace.

Add a Router:

Drag and drop a Router (e.g., 1941 model) from *Network Devices* to the workspace for inter-VLAN communication.

3. Cable the Network

Use a Copper Straight-Through Cable to connect:

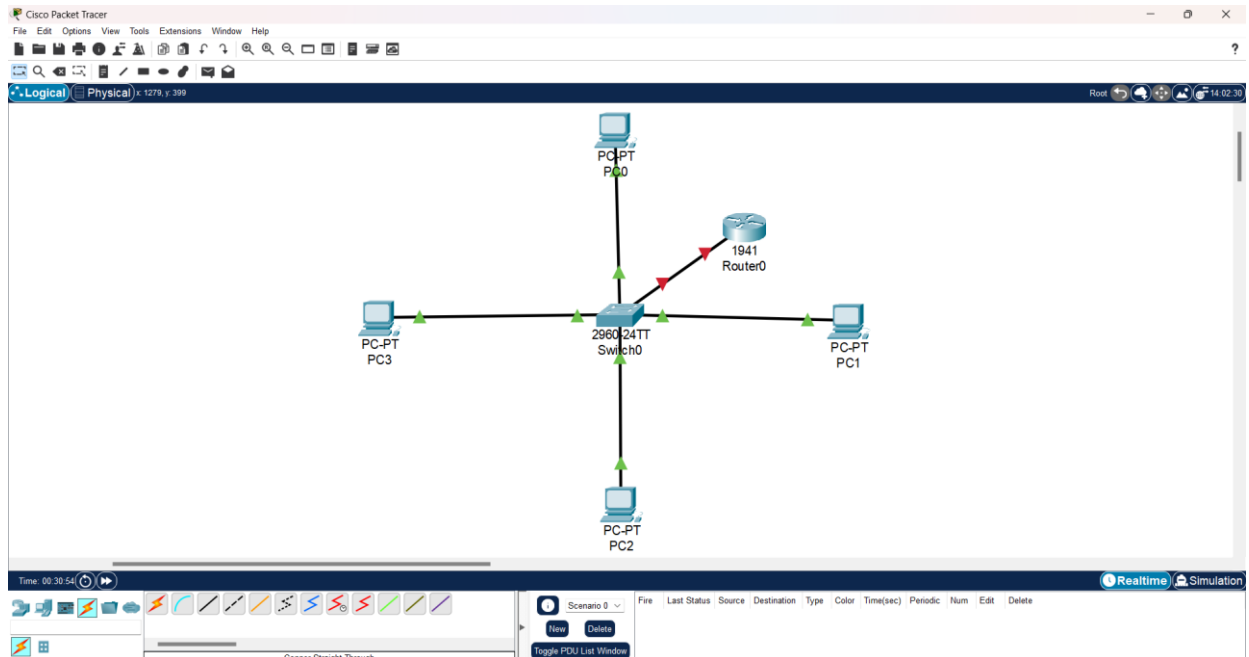
PC1 to FastEthernet0/1 of the switch.

PC2 to FastEthernet0/2 of the switch.

PC3 to FastEthernet0/3 of the switch.

PC4 to FastEthernet0/4 of the switch.

Switch to the Router's GigabitEthernet0/0 interface using a Copper Straight-Through Cable.



4. VLAN Creation on the Switch

Access the Switch CLI:

Click on the switch and go to the CLI tab.

```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Motherboard serial number : FOC10093R12
Power supply serial number : AZS1007032H
Model revision number : B0
Motherboard revision number : B0
Model number : WS-C2960-24TT-L
System serial number : FOC1010X104
Top Assembly Part Number : 800-27221-02
Top Assembly Revision Number : A0
Version ID : V02
CLEI Code Number : COM3L00BRA
Hardware Board Revision Number : 0x01

Switch Ports Model          SW Version  SW Image
-----
* 1 26 WS-C2960-24TT-L 15.0(2)SE4 C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnnguyen

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
```

Enter Configuration Mode:

CODE:

enable

configure terminal

Create VLANs:**Create VLAN 10 and VLAN 20:**

CODE:

```
vlan 10
name Sales
exit
vlan 20
name Marketing
exit
```

Assign VLANs to Ports:**Assign FastEthernet0/1 and FastEthernet0/2 to VLAN 10:**

CODE:

```
interface fastethernet0/1
switchport mode access
switchport access vlan 10
exit
interface fastethernet0/2
switchport mode access
switchport access vlan 10
exit
```

Assign FastEthernet0/3 and FastEthernet0/4 to VLAN 20:

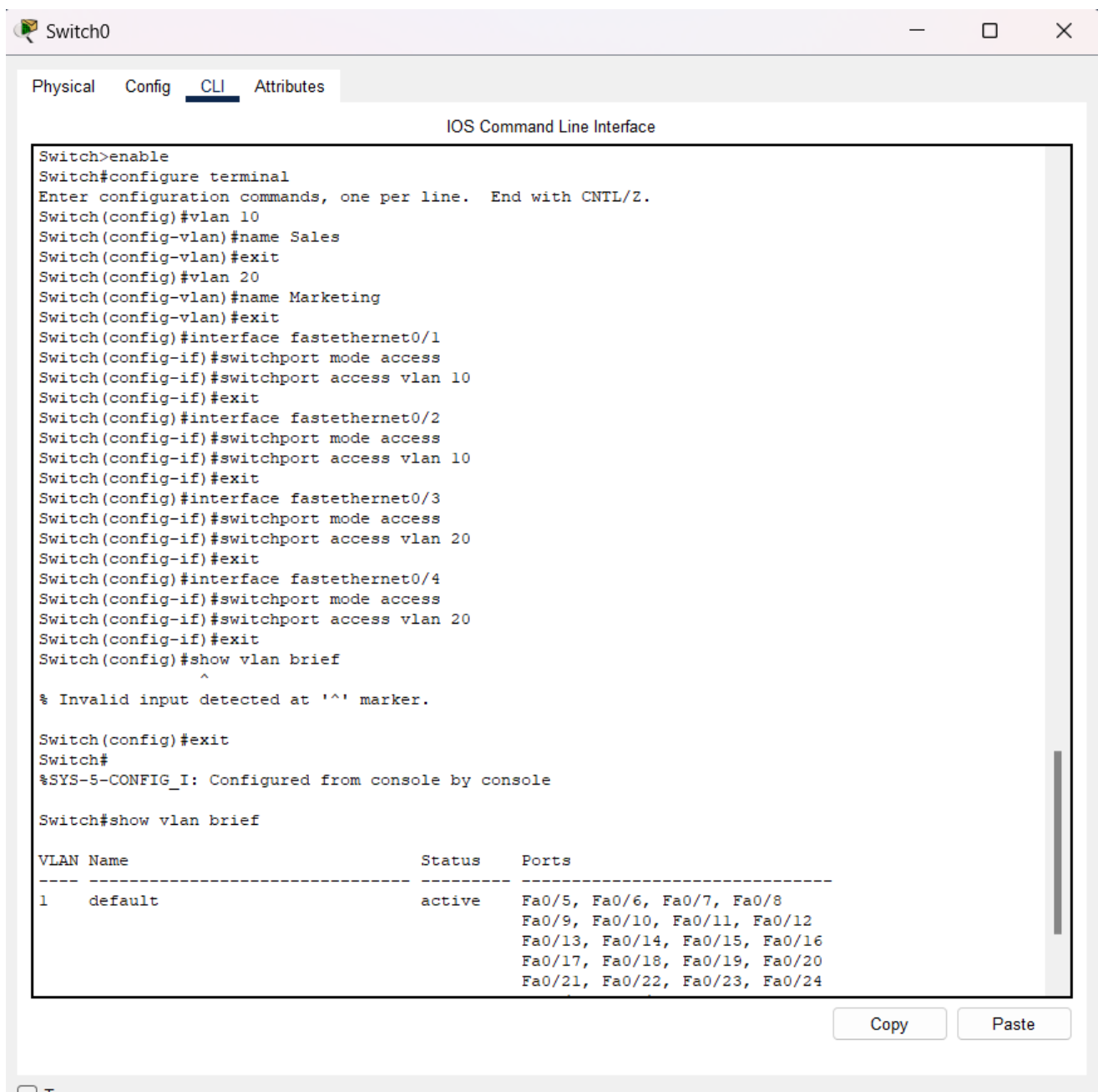
CODE:

```
interface fastethernet0/3
switchport mode access
switchport access vlan 20
exit
interface fastethernet0/4
switchport mode access
switchport access vlan 20
exit
```

Verify VLAN Configuration:

CODE:

```
show vlan brief
```



5. Configure the Router for Inter-VLAN Communication

Assign Subinterfaces:

Access the router CLI and configure GigabitEthernet0/0:

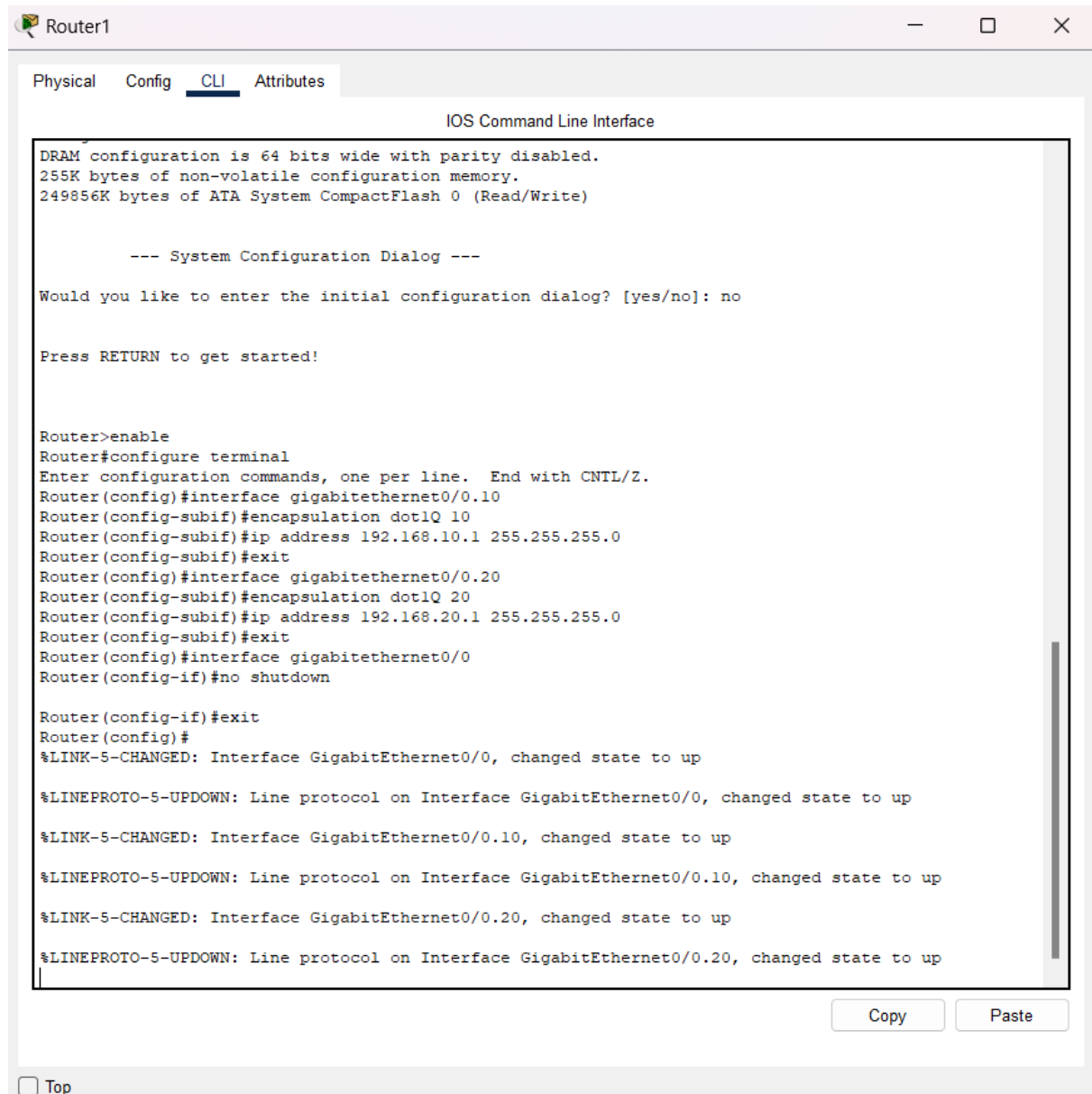
CODE:

```
enable
configure terminal
interface gigabitethernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit
interface gigabitethernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
```


Activate the Router Interface:

CODE:

```
interface gigabitethernet0/0
no shutdown
exit
```



6. Configure IP Addresses on PCs

Assign static IP addresses to each PC as follows:

PC1: 192.168.10.2, Subnet Mask: 255.255.255.0, Gateway: 192.168.10.1

PC2: 192.168.10.3, Subnet Mask: 255.255.255.0, Gateway: 192.168.10.1

PC3: 192.168.20.2, Subnet Mask: 255.255.255.0, Gateway: 192.168.20.1

PC4: 192.168.20.3, Subnet Mask: 255.255.255.0, Gateway: 192.168.20.1

On each PC:

Open the Desktop tab.

Select IP Configuration.

Input the above IP details.

7. Test Communication

Ping PCs in the Same VLAN:

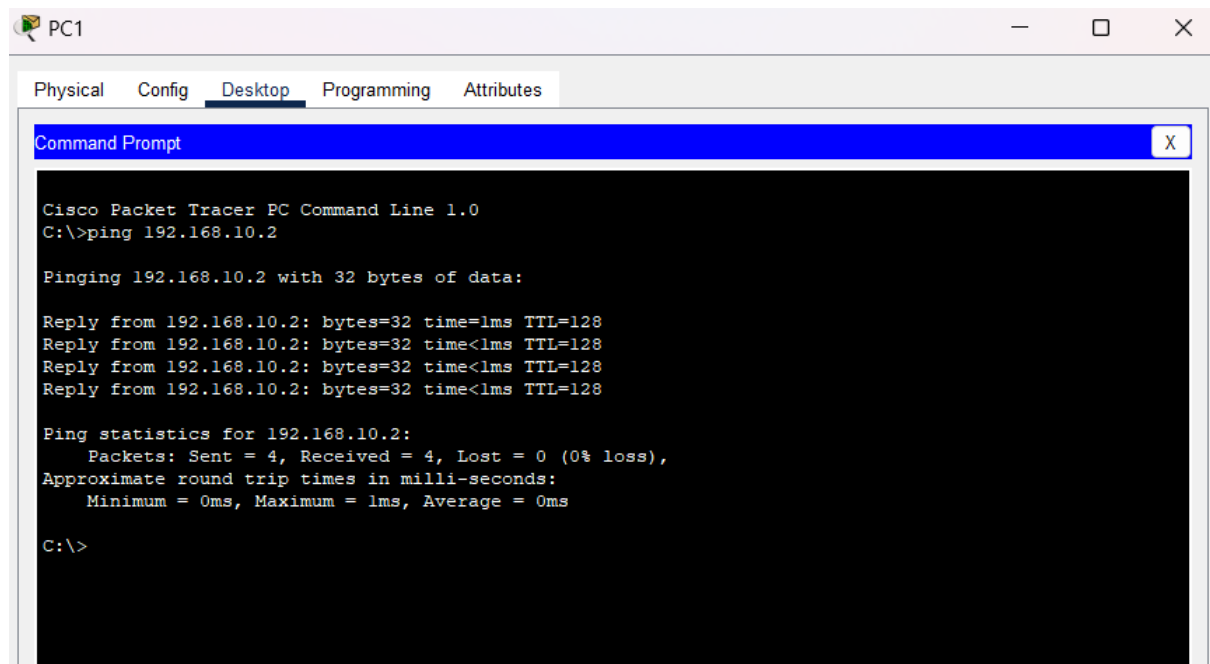
From PC1, ping PC2 (successful).

From PC3, ping PC4 (successful).

Ping PCs Across VLANs:

From PC1, ping PC3 (successful).

From PC2, ping PC4 (successful).



The screenshot shows a window titled 'PC1' with a 'Desktop' tab selected. Inside the window is a 'Command Prompt' titled 'Cisco Packet Tracer PC Command Line 1.0'. The command prompt shows the command 'C:\>ping 192.168.10.2' being executed. The output shows four successful replies from 192.168.10.2 with 32 bytes of data, each taking less than 1ms and having a TTL of 128. The ping statistics show 4 packets sent, 4 received, and 0% loss, with round trip times of 0ms, 1ms, and 0ms respectively.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

8. Outcome:

VLANs were successfully created and configured on the switch.

Communication within the same VLAN was successful.

Inter-VLAN communication was facilitated using a router.

Students learned practical VLAN configuration and inter-VLAN routing.

Practical 4

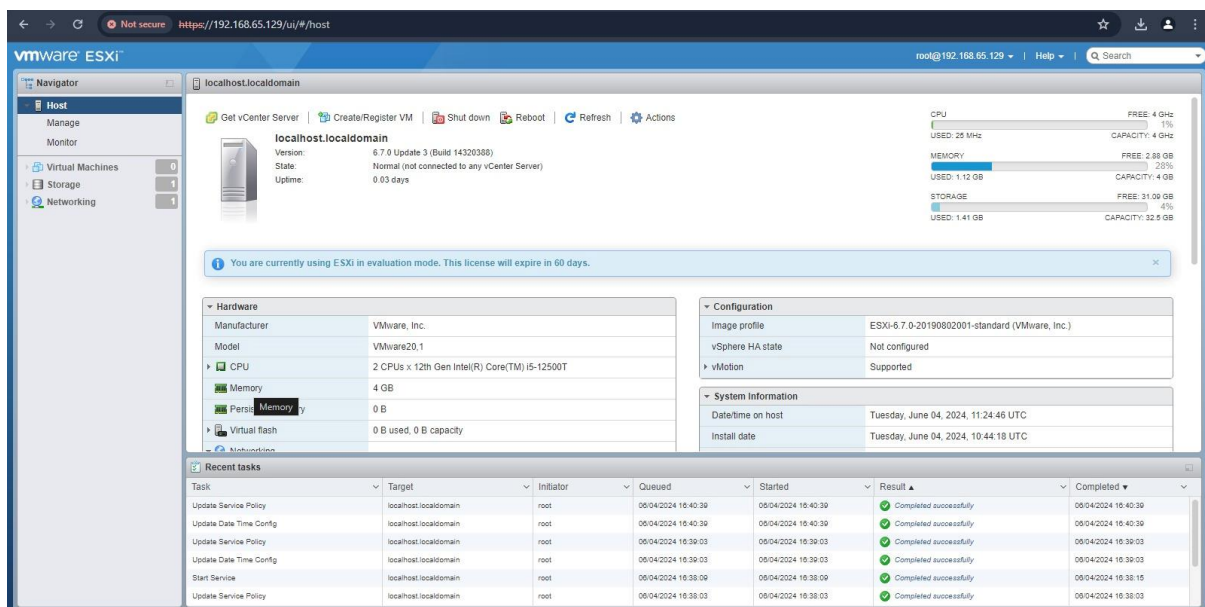
Creating a Virtual Machine

Theory: - Creating a virtual machine (VM) involves using virtualization software to emulate a computer system within another computer system, allowing multiple operating systems (OS) to run simultaneously on a single physical machine. The process typically begins with selecting a virtualization platform such as VMware, VirtualBox, or Hyper-V, and then configuring parameters such as CPU, memory, storage, and networking for the virtual machine. Next, an installation image of the desired operating system is obtained and loaded into the virtualization software, which creates a virtual environment where the OS can operate independently. Once the VM is created, it can be started, stopped, cloned, or deleted as needed, providing a flexible and efficient way to deploy and manage computing resources for various purposes, such as development, testing, or running legacy applications.

Step 1: Create a New VM

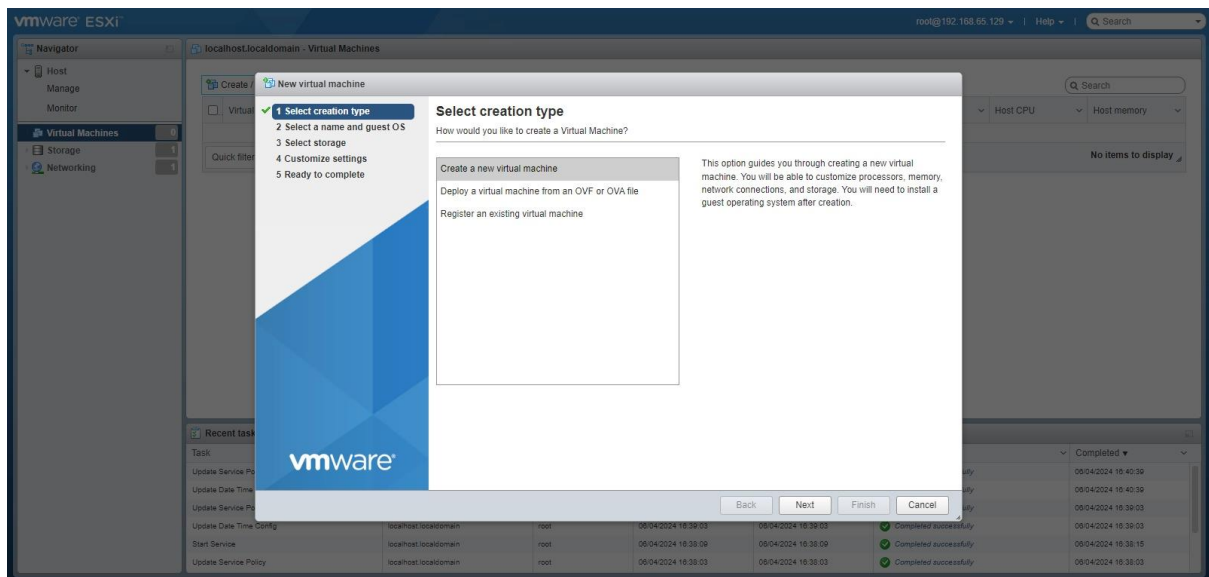
Open the vSphere Host Client

1. Open a web browser on your computer.
2. Navigate to 'https://[Your-ESXi-Host-IP-Address]'`.
3. Log in with your root account and password.



Create/Register a New VM

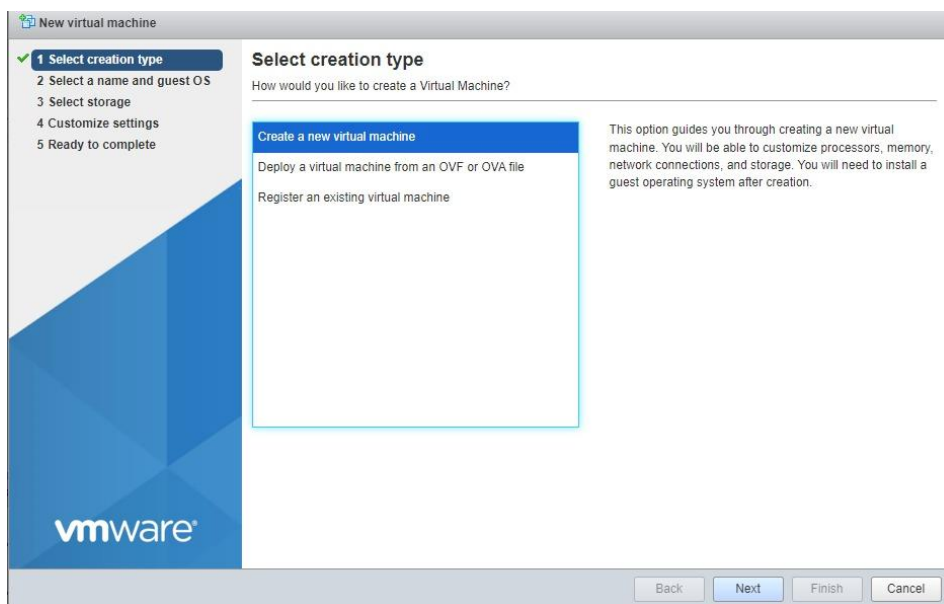
1. Once logged in, go to the **Navigator** panel on the left.
2. Click on **Virtual Machines**.
3. Click the **Create / Register VM** button.



Follow the Wizard

1. Select creation type:

- Choose **Create a new virtual machine**.
- Click **Next**.



2. Select a name and guest OS:

- Enter a name for your VM.
- Select the appropriate **Compatibility** (default is usually fine).
- Choose a **Guest OS family** (e.g., Linux, Windows).
- Select the specific **Guest OS version**.
- Click **Next**.

New virtual machine - MY OS (ESXi 6.7 virtual machine)

1 Select creation type
2 Select a name and guest OS
 3 Select storage
 4 Customize settings
 5 Ready to complete

Select a name and guest OS

Specify a unique name and OS

Name
 MY OS

Virtual machine names can contain up to 80 characters and they must be unique within each ESXi instance.

Identifying the guest operating system here allows the wizard to provide the appropriate defaults for the operating system installation.

Compatibility
 ESXi 6.7 virtual machine

Guest OS family
 Windows

Guest OS version
 Microsoft Windows 10 (64-bit)

☐ Enable Windows Virtualization Based Security 

Back Next Finish Cancel

3. Select storage:

- Choose the datastore where you want to store the VM files.
- Click **Next**.

New virtual machine - MY OS (ESXi 6.7 virtual machine)

1 Select creation type
 2 Select a name and guest OS
3 Select storage
 4 Customize settings
 5 Ready to complete

Select storage

Select the storage type and datastore

Standard Persistent Memory

Select a datastore for the virtual machine's configuration files and all of its' virtual disks.

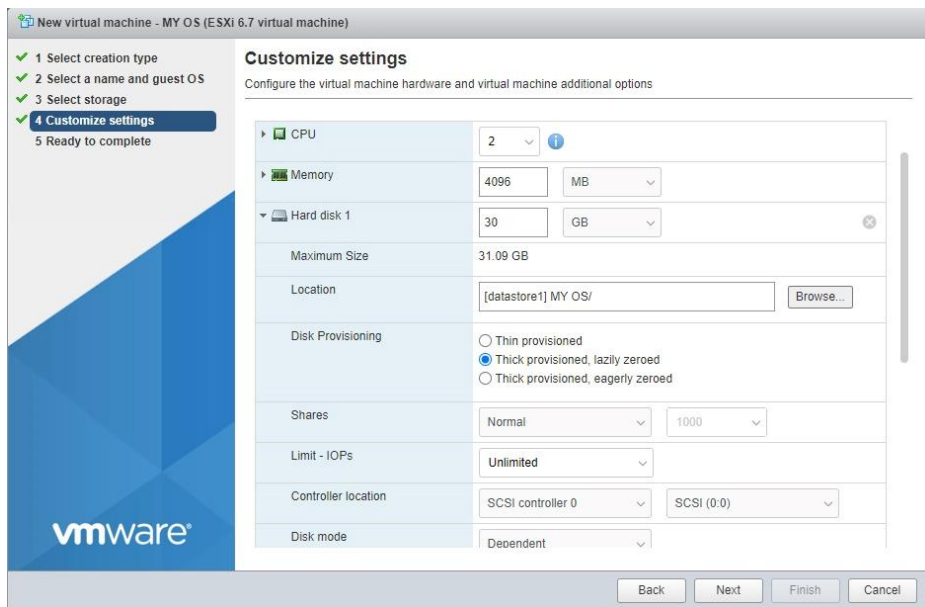
Name	Capacity	Free	Type	Thin pro...	Access
datastore1	32.5 GB	31.09 GB	VMFS6	Supported	Single

1 items

Back Next Finish Cancel

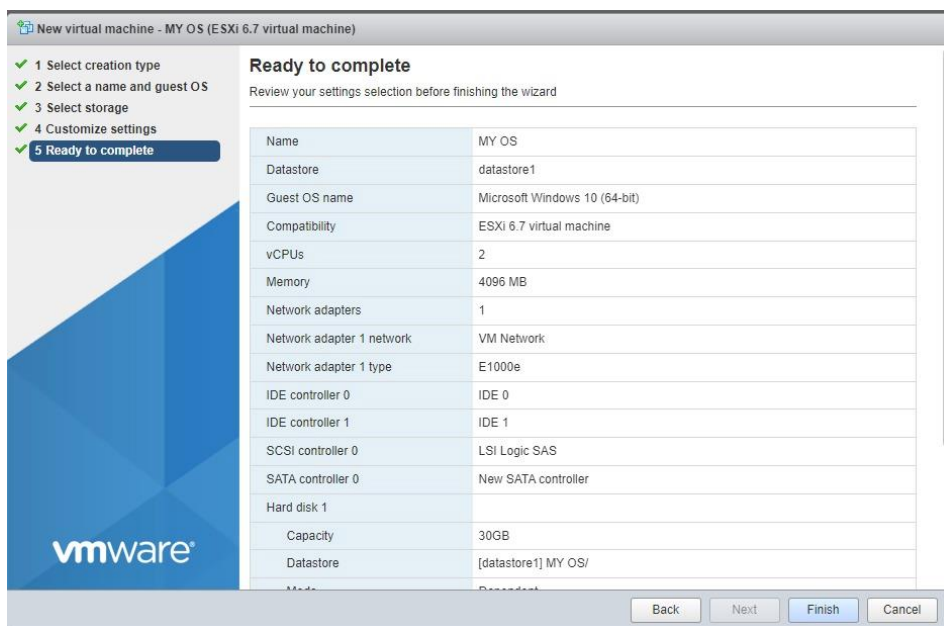
4. Customize settings:

- Configure the number of **CPUs** and **cores** per CPU.
- Set the amount of **Memory** (RAM) for the VM.
- Adjust the **Disk provisioning** (e.g., Thick Provision Lazy Zeroed, Thin Provision).
- Specify the **Disk size**.
- Optionally, add or remove any hardware components (e.g., network adapters, USB controllers).
- Click **Next**.



5. Ready to complete:

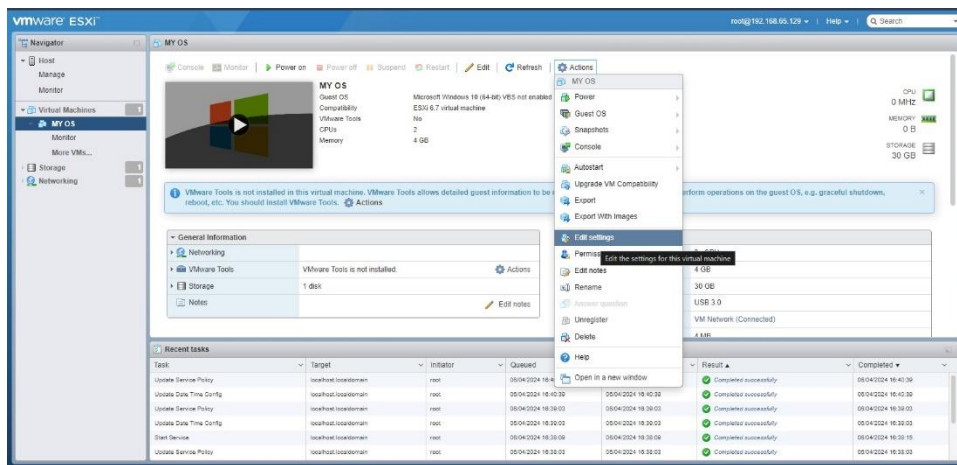
- Review your settings.
- Click **Finish** to create the VM.



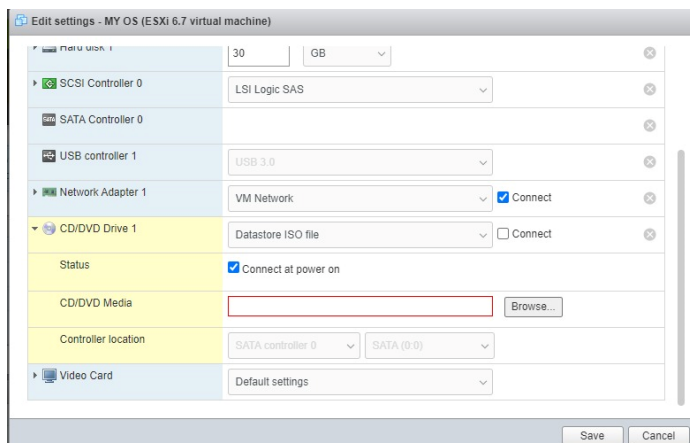
Step 2: Install Guest OS

Attach an ISO Image of the Guest OS to the VM

1. In the vSphere Host Client, click on the **Virtual Machines** tab.
2. Select the newly created VM from the list.
3. Click on **Actions > Edit Settings**.

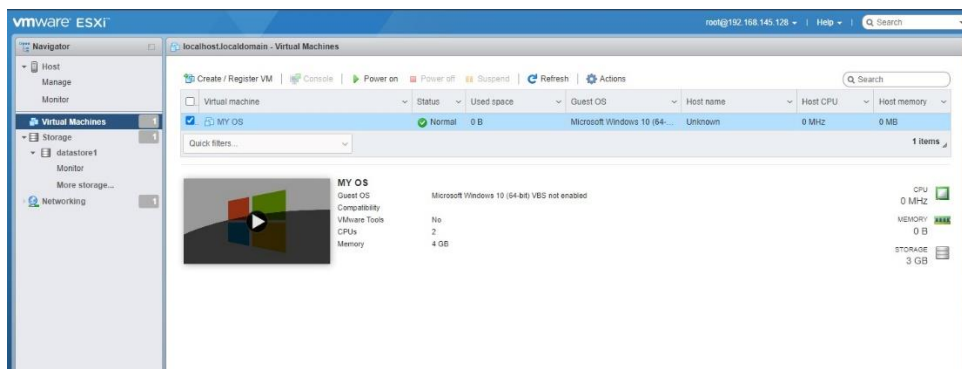


4. In the **Virtual Hardware** tab, locate the **CD/DVD Drive 1**.
5. Set the **Device Type** to **Datastore ISO File**.
6. Click on **Browse** and select the ISO image of the guest OS you want to install.
7. Ensure **Connect at power on** is checked.
8. Click **Save**.

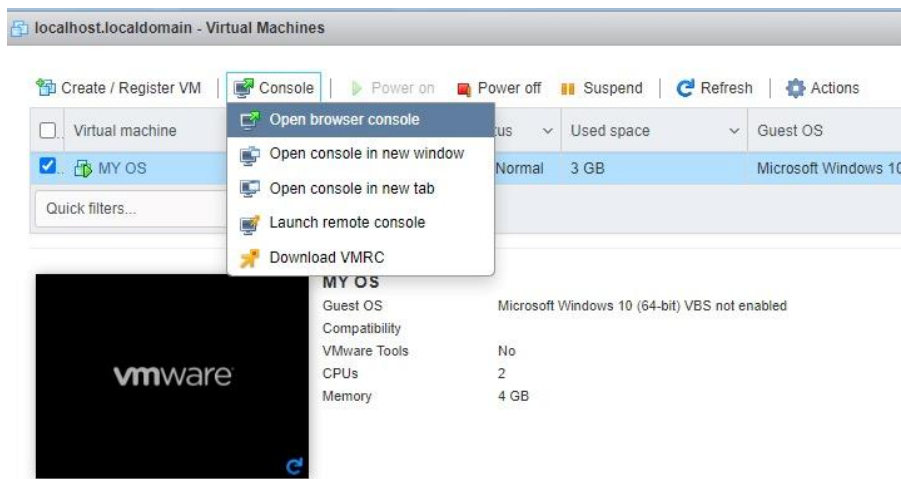


Start the VM and Follow the OS Installation Steps

1. With the VM selected, click on the **Power On** button.



2. Click on **Launch Web Console** to open the console window.



3. The VM should boot from the attached ISO image.

4. Follow the on-screen prompts to install the guest OS.

- This typically involves selecting the installation language, configuring the disk where the OS will be installed, and entering any required product keys or configuration settings.

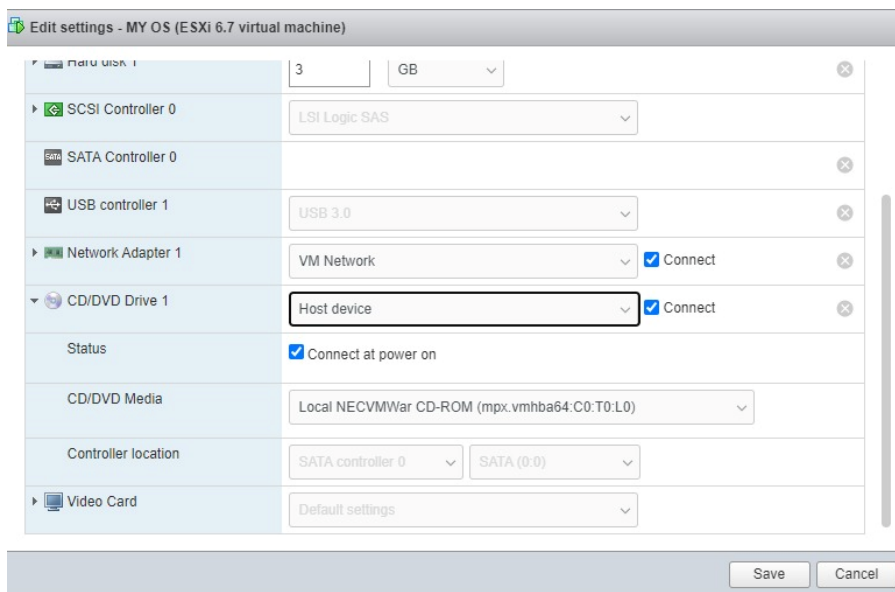
5. Once the installation is complete, the VM will typically reboot.

6. Remove the ISO image from the CD/DVD drive to avoid booting from it again:

- Go to **Actions > Edit Settings**.

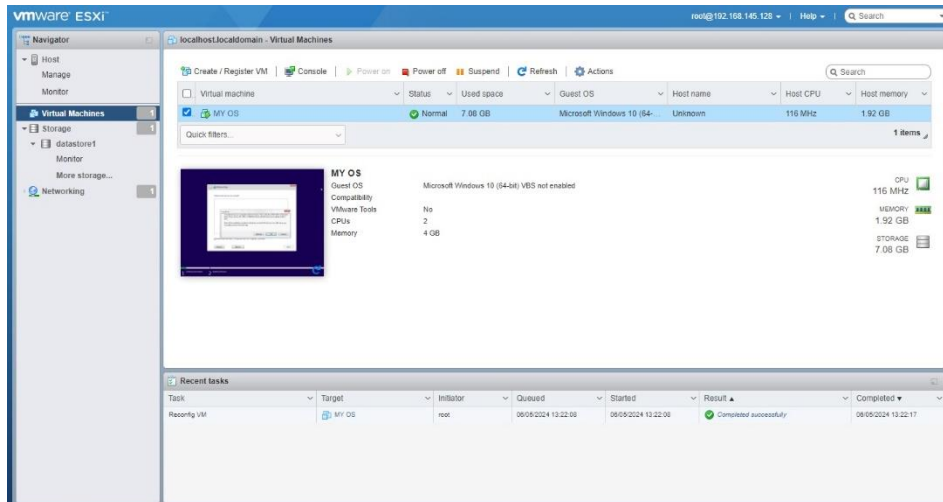
- Change the **Device Type** of the CD/DVD Drive to **Client Device** or **Host Device** (depending on your preference).

- Click **Save**.



Final Steps

1. Ensure that VMware Tools (or Open VM Tools for Linux VMs) is installed on the guest OS to enhance performance and manageability.
2. Update the guest OS with the latest patches and updates.



You have now successfully created a virtual machine on VMware ESXi and installed a guest operating system on it.

Practical 5

Configure a Basic Load Balancer

Objective: Simulate load balancing in a lab setup using HAProxy to distribute requests between two web servers.

Steps to Implement

Step 1: Set up the Environment

1. Prepare a Linux Machine:

Ensure you have a Linux system (e.g., Ubuntu, CentOS).

- **Update system packages:**

Code:

```
sudo apt update && sudo apt upgrade -y # For Ubuntu
```

```
sudo yum update -y # For CentOS
```

```
vboxuser@ubuntu1:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for vboxuser:
Get:1 file:/cdrom/noble InRelease
Ign:1 file:/cdrom/noble InRelease
Get:2 file:/cdrom/noble Release
Err:2 file:/cdrom/noble Release
  File not found - /cdrom/dists/noble/Release (2: No such file or directory)
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu noble InRelease
Get:5 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:6 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
E: The repository 'file:/cdrom/noble Release' no longer has a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
vboxuser@ubuntu1:~$
```

2. Install HAProxy:

Code:

```
sudo apt install haproxy -y # For Ubuntu
```

```
sudo yum install haproxy -y # For CentOS
```

```
vboxuser@ubuntu1:~$ sudo apt install haproxy -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  vim-haproxy haproxy-doc
The following NEW packages will be installed:
  haproxy
0 upgraded, 1 newly installed, 0 to remove and 3 not upgraded.
Need to get 2,038 kB of archives.
After this operation, 4,631 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 haproxy amd64 2.8.5-1ubuntu3.1 [2,038 kB]
Fetched 2,038 kB in 4s (531 kB/s)
Selecting previously unselected package haproxy.
(Reading database ... 152091 files and directories currently installed.)
Preparing to unpack .../haproxy_2.8.5-1ubuntu3.1_amd64.deb ...
Unpacking haproxy (2.8.5-1ubuntu3.1) ...
Setting up haproxy (2.8.5-1ubuntu3.1) ...
invoke-rc.d: policy-rc.d denied execution of start.
Created symlink /etc/systemd/system/multi-user.target.wants/haproxy.service → /usr/lib/systemd/system/haproxy.service.
/usr/sbin/policy-rc.d returned 101, not running 'start haproxy.service'
Processing triggers for rsyslog (8.2312.0-3ubuntu9) ...
invoke-rc.d: policy-rc.d denied execution of try-restart.
Processing triggers for man-db (2.12.0-4build2) ...
vboxuser@ubuntu1:~$
```

Step 2: Set up Web Servers

1. Install Python (if not already installed):

Code:

```
sudo apt install python3 -y # For Ubuntu
sudo yum install python3 -y # For CentOS
```

```
vboxuser@ubuntu1:~$ sudo apt install python3 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.12.3-0ubuntu2).
python3 set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
vboxuser@ubuntu1:~$
```

2. Create Two Simple HTTP Servers:

o Server 1:

Open a terminal and run:

Code:

```
mkdir ~/server1 && cd ~/server1
echo "Hello from Server 1" > index.html
python3 -m http.server 8081
```

```
vboxuser@ubuntu1:~$ mkdir ~/server1 && cd ~/server1
vboxuser@ubuntu1:~/server1$ echo "Hello from Server 1" > index.html
vboxuser@ubuntu1:~/server1$ python3 -m http.server 8081
Serving HTTP on 0.0.0.0 port 8081 (http://0.0.0.0:8081/) ...
```

- **Server 2:**
Open another terminal and run:

Code:

```
mkdir ~/server2 && cd ~/server2
echo "Hello from Server 2" > index.html
python3 -m http.server 8082
```

```
vboxuser@ubuntu1:~$ mkdir ~/server2 && cd ~/server2
vboxuser@ubuntu1:~/server2$ echo "Hello from Server 2" > index.html
vboxuser@ubuntu1:~/server2$ python3 -m http.server 8082
Serving HTTP on 0.0.0.0 port 8082 (http://0.0.0.0:8082/) ...
```

Step 3: Configure HAProxy

- 1. Edit the HAProxy Configuration File:**
Open the configuration file:

Code:

```
sudo nano /etc/haproxy/haproxy.cfg
```

Add the following configuration:

Code:

```
global
    log /dev/log local0
    log /dev/log local1 notice
    daemon

defaults
    log global
    option httplog
    timeout connect 5000ms
    timeout client 50000ms
    timeout server 50000ms

frontend http_front
    bind *:80
    default_backend http_back

backend http_back
    balance roundrobin
    server server1 127.0.0.1:8081 check
    server server2 127.0.0.1:8082 check
```

2. Restart HAProxy Service:

Save the file and restart HAProxy:

Code:

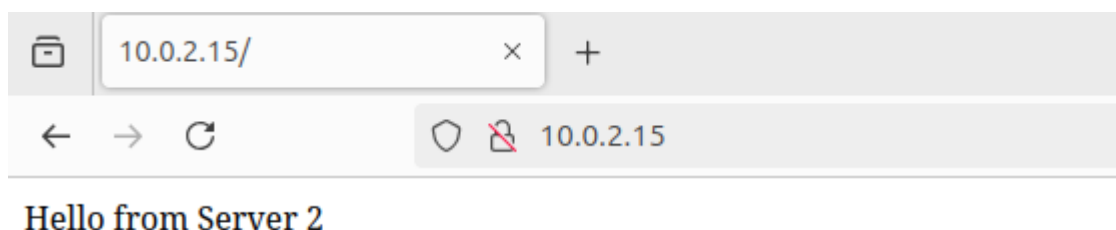
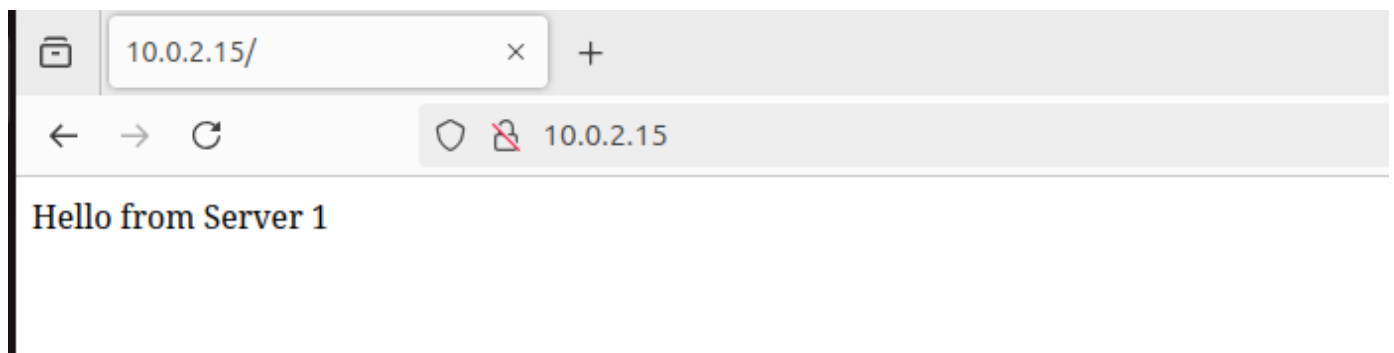
```
sudo systemctl restart haproxy
```

```
sudo systemctl enable haproxy
```

```
vboxuser@Ubuntu:~$ sudo systemctl restart haproxy
[sudo] password for vboxuser:
vboxuser@Ubuntu:~$ sudo systemctl enable haproxy
Synchronizing state of haproxy.service with SysV service script with /usr/lib/sy
stemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable haproxy
vboxuser@Ubuntu:~$
```

Step 4: Test the Load Balancer

1. Open a web browser and navigate to <http://<Linux-machine-IP>>.
2. Refresh the page multiple times to observe requests being distributed between the two servers.
 - You should alternately see "Hello from Server 1" and "Hello from Server 2."



Outcome

You have successfully configured a basic load balancer using HAProxy. This practical exercise demonstrates how load balancing can distribute traffic evenly across multiple servers, ensuring better resource utilization and availability.

Practical 6

Configure Virtual PortChannels

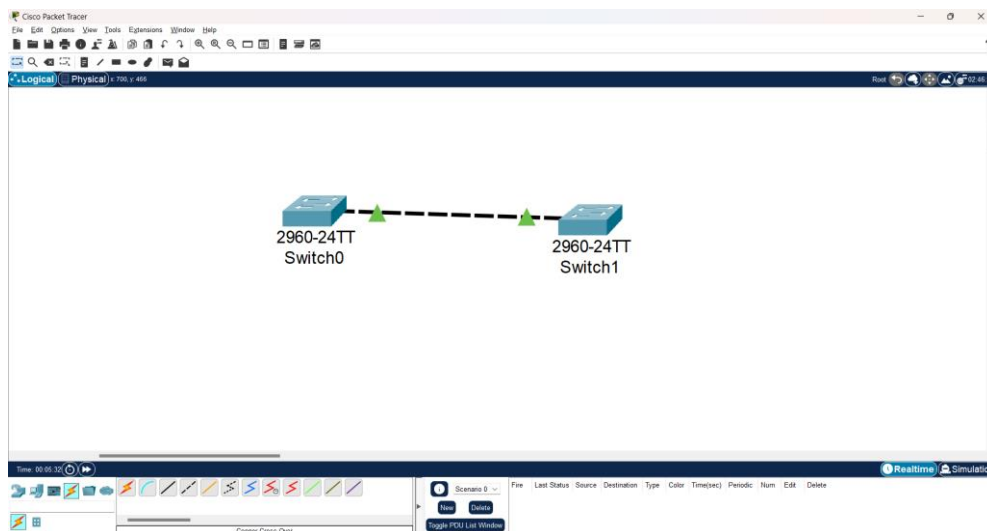
Objective:

Simulate the use of PortChannels for link aggregation to increase bandwidth and provide failover.

Steps:

1. Setup the Topology in Packet Tracer:

- Add two switches (e.g., Switch1 and Switch2) to the workspace.
- Connect the switches with **two physical links** using Ethernet cables (Copper Cross-Over Cabel) (e.g., GigabitEthernet0/1 and GigabitEthernet0/2).



2. Configure the PortChannels on Switch1:

- Access the CLI of Switch1.
- Enter global configuration mode and create a PortChannel interface:

Code:

```
configure terminal
```

```
interface range GigabitEthernet0/1-2
```

```
channel-group 1 mode active
```


exit

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range GigabitEthernet0/1-2
Switch(config-if-range)#channel-group 1 mode active
Switch(config-if-range)#exit
Switch(config)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

- Verify the PortChannel interface:

Code:

show etherchannel summary

```
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Pol(SD)          LACP       Gig0/1(I) Gig0/2(D)
Switch#
```

3. Configure the PortChannels on Switch2:

- Access the CLI of Switch2.
- Repeat the steps as on Switch1:

Code:

configure terminal

interface range GigabitEthernet0/1-2

channel-group 1 mode active

exit

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range GigabitEthernet0/1-2
Switch(config-if-range)#channel-group 1 mode active
Switch(config-if-range)#exit
Switch(config)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

```

- Verify the PortChannel interface:

Code:

show etherchannel summary

```

Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Pol(SU)          LACP       Gig0/1(P) Gig0/2(D)
Switch#

```

4. Test Bandwidth and Failover:

- Enable both links and check the PortChannel's aggregated bandwidth using:

Code:

show interfaces port-channel 1


```

Switch#show interfaces port-channel 1
Port-channel1 is up, line protocol is up (connected)
  Hardware is EtherChannel, address is 0090.0c5a.659b (bia 0090.0c5a.659b)
  MTU 1500 bytes, BW 2000000 Kbit, DLY 1000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 2000Mb/s
  input flow-control is off, output flow-control is off
  Members in this channel: Gig0/1 ,
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
--More-- |

```

- Simulate a link failure by disabling one link (e.g., GigabitEthernet0/1) on Switch1:

Code:

```
interface GigabitEthernet0/1
```

```
shutdown
```

```
exit
```

```

Switch#enable
Switch#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface GigabitEthernet0/1
Switch(config-if)#shutdown

Switch(config-if)#exit
Switch(config)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINK-3-UPDOWN: Interface Port-channel1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down
|

```

- Verify that traffic continues to flow through the remaining link:

Code:

```
show etherchannel summary
```

```
Switch#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby  (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
```

```
Number of channel-groups in use: 1
Number of aggregators:          1
```

Group	Port-channel	Protocol	Ports
1	Pol(SD)	LACP	Gig0/1(D) Gig0/2(D)

```
Switch#
```

- Reactivate the link:

Code:

```
interface GigabitEthernet0/1
```

```
no shutdown
```

```
exit
```

```
Switch#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface GigabitEthernet0/1
Switch(config-if)#no shutdown

Switch(config-if)#exit
Switch(config)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up
```

Outcome:

- Successfully combined two physical links into a single logical PortChannel to:
 1. Increase bandwidth by aggregating the links.
 2. Ensure failover capability where traffic switches to the operational link if one link fails.
- Verified the configuration and observed the benefits of link aggregation in network resiliency and performance.

Practical 7

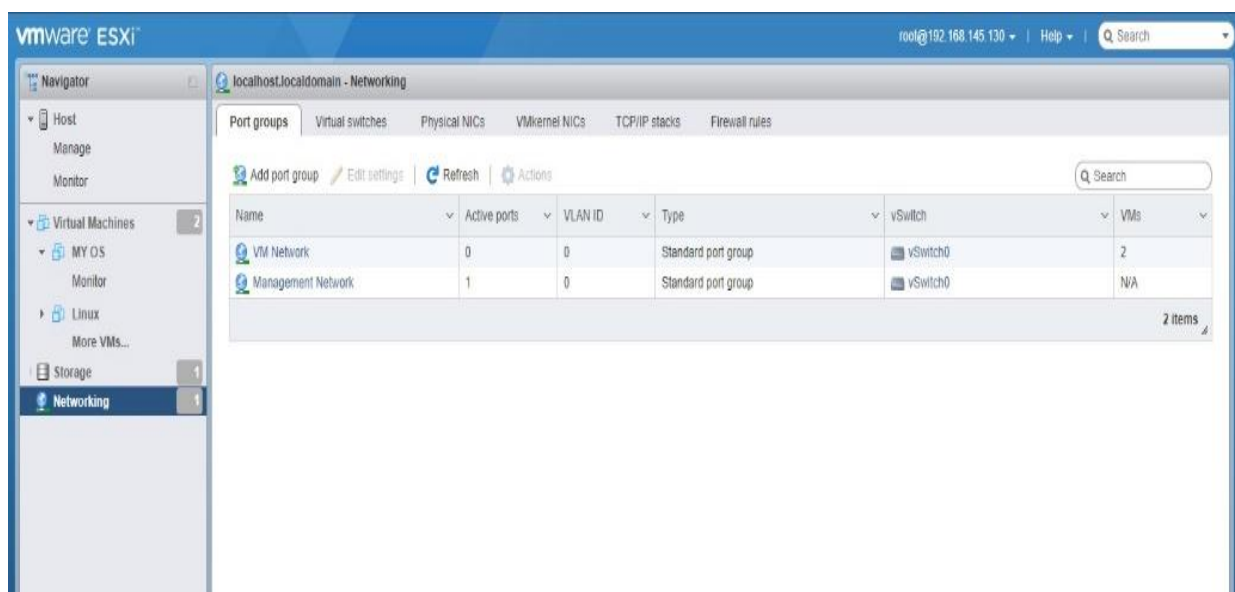
Creating and Managing VM Networks

Theory: - Creating and managing virtual machine (VM) networks involves configuring virtual networks within a hypervisor environment to enable communication between virtual machines and external networks. This process typically begins with defining network adapters and configuring network settings such as IP addresses, subnet masks, gateways, and DNS servers for each VM. Virtual switches, routers, and firewalls can be implemented to control network traffic and ensure security. Network segmentation and VLANs may be used to isolate traffic between different VMs or groups of VMs. Monitoring tools and management interfaces allow administrators to oversee network performance, troubleshoot connectivity issues, and allocate resources effectively. By efficiently managing VM networks, administrators can optimize performance, enhance security, and facilitate seamless communication between virtual machines and external resources.

Step 1: Create a Virtual Switch

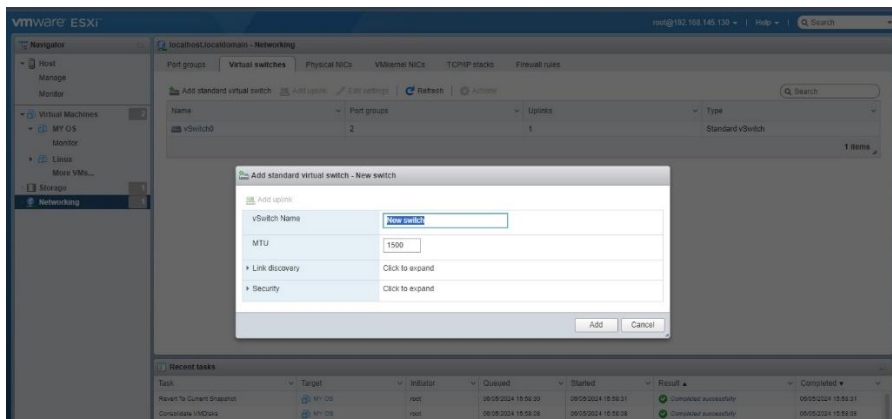
Go to "Networking"

1. Open a web browser and navigate to `https://[Your-ESXi-Host-IP-Address]`.
2. Log in with your root account and password.
3. In the **Navigator** panel, click on **Networking**.



Choose "Add Networking" and Create a New Virtual Switch

1. In the **Networking** section, click on **Virtual switches**.
2. Click on **Add standard virtual switch** (or **Add networking**, depending on the interface version).



3. In the **Add Networking Wizard**:

- **Select target device:** Choose **Create a virtual switch** and click **Next**.
- **Name and VLAN:** Enter a name for the new virtual switch (e.g., vSwitch1).
- Optionally, configure VLAN settings if you are using VLANs in your network.
- **NIC Teaming:** Assign physical network adapters (uplinks) to the virtual switch if needed. This is optional and depends on your networking requirements.
- **Review:** Review your settings and click **Finish** to create the virtual switch.

Add standard virtual switch - New switch

Add uplink

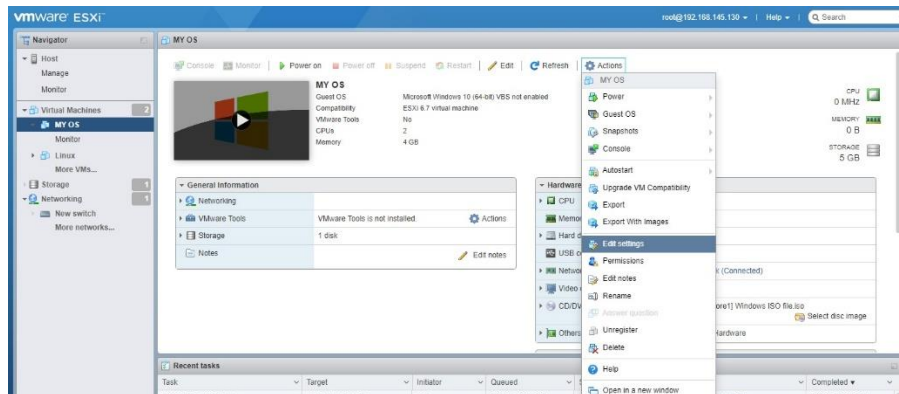
vSwitch Name	New switch
MTU	1500
▼ Link discovery	
Mode	Listen
Protocol	Cisco discovery protocol (CDP)
▼ Security	
Promiscuous mode	☐ Accept ☒ Reject
MAC address changes	☐ Accept ☒ Reject
Forged transmits	☐ Accept ☒ Reject

Add Cancel

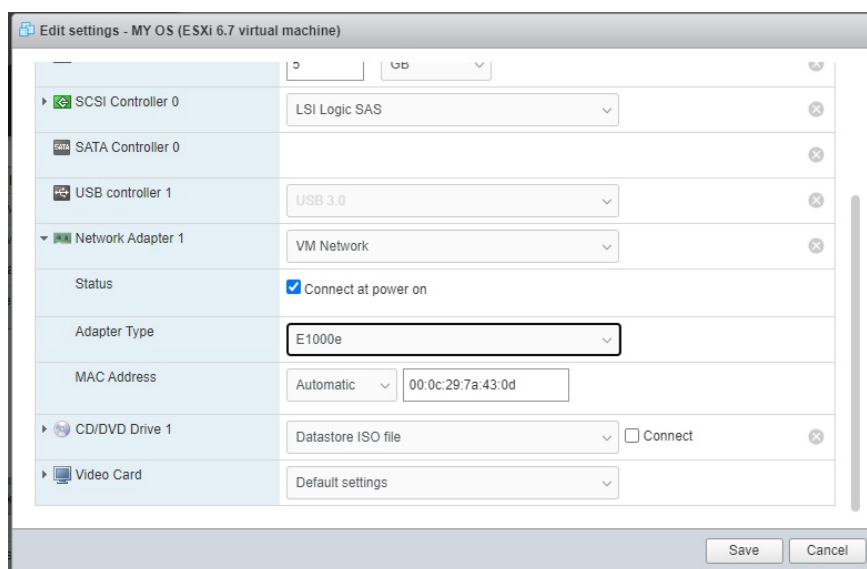
Step 2: Assign VM to Virtual Switch

Edit the VM Settings and Attach its Network Adapter to the Newly Created Virtual Switch

1. In the **Navigator** panel, click on **Virtual Machines**.
2. Select the VM you want to assign to the new virtual switch.
3. With the VM selected, click on **Actions > Edit Settings**.



4. In the **Edit Settings** dialog:
 - o **Network Adapter:** Locate the **Network adapter** section.
 - o **Network Label:** Change the **Network Label** to the name of the newly created virtual switch (e.g., vSwitch1).
5. Click **Save** to apply the changes.



By following these steps, you have successfully created a virtual switch and assigned a VM to it. This allows you to manage and segment your network traffic within the ESXi host.

Practical 8

Spanning Tree Protocol (STP) Simulation Guide

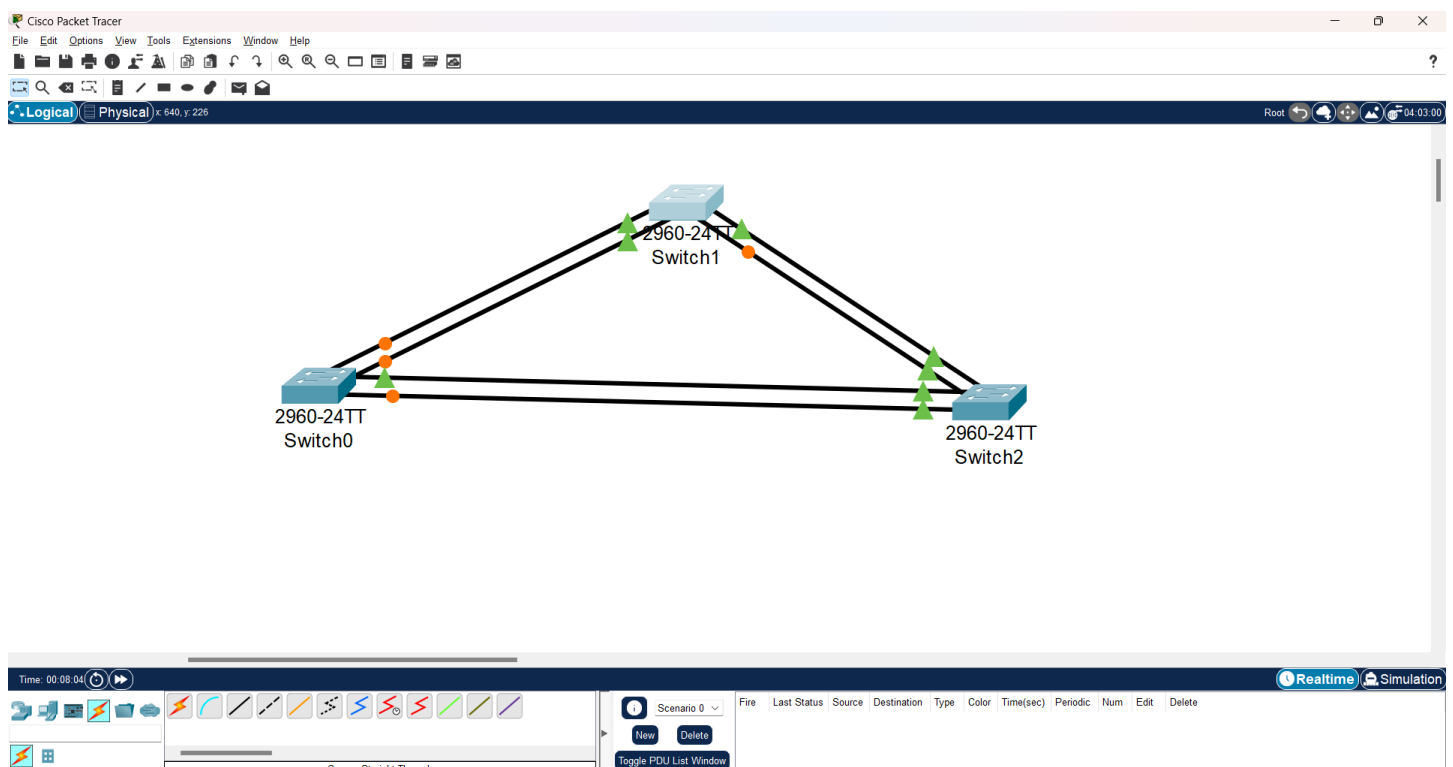
Objective:

Prevent network loops using Spanning Tree Protocol (STP) to ensure network stability.

Steps:

1. Create a Network with Redundant Links

- **Open Cisco Packet Tracer.**
- **Add at least three switches** to the workspace (e.g., Switch0, Switch1, and Switch2).
- **Interconnect the switches with redundant links.** For instance:
 - Connect **Switch0** to **Switch1** with two links.
 - Connect **Switch1** to **Switch2** with two links.
 - Connect **Switch0** to **Switch2** with two links.
- Use **crossover cables** for inter-switch connections if using older switches; otherwise, use straight-through cables.



2. Enable STP on the Switches

- **Configure the switches:**
 1. Go to the CLI (Command Line Interface) of each switch (repeat the same step for each switch).

2. Enter the following commands:

Code:

```
enable
configure terminal
spanning-tree vlan 1
exit
```

```
Switch>enable
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#spanning-tree vlan 1
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

- The spanning-tree vlan 1 command enables STP for VLAN 1 (default VLAN).

• Verify STP operation:

- Use the show spanning-tree command to observe the status of STP. The output should show:
 - The **Root Bridge** (the switch elected as the root).
 - The **ports in forwarding state**.
 - The **ports in blocking state** (redundant links).

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.C90E.99A5
             Cost        19
             Port        3(FastEthernet0/3)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     000A.41AE.17B1
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/4	Altn	BLK	19	128.4	P2p
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/3	Root	FWD	19	128.3	P2p

```
Switch#
```

3. Observe STP in Action

- Once STP is enabled:
 - Check which ports are **blocked** to prevent loops.

- The show spanning-tree command will display which interfaces are in **Forwarding** or **Blocking** states.

4. Simulate a Link Failure

- Disconnect one of the **primary links** between two switches. For example:
 - Disconnect a link between Switch0 and Switch1.
- Observe how STP reacts:
 - Use the show spanning-tree command to verify that a **blocked link** is now in the **forwarding state**, restoring network connectivity.

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.C90E.99A5
             Cost        19
             Port        3(FastEthernet0/3)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     000A.41AE.17B1
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/4	Altn	FWD	19	128.4	P2p
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/3	Root	FWD	19	128.3	P2p

```
Switch#
```

5. Verify Network Stability

- Use **ping** to ensure connectivity between devices on the network remains intact during the simulation.
- Verify that STP successfully prevented any loops.

Outcome:

By completing this simulation, you will:

- Understand how STP blocks redundant links to prevent network loops.
- See STP's role in automatically reactivating links to maintain network stability when a primary link fails.
- Gain practical knowledge of configuring and monitoring STP in a network environment.